Galahad and the Siege Perilous: A Cheat-Proof Trial for Language Grounding, and the First Policy to Survive It

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Φ(fight) Research†

Abstract

A vision-language-action policy can score 97% on a manipulation benchmark while ignoring its instruction: replace the words with the string "xxx" and it emits the same actions. It succeeds by walking to the object that usually sits in that position, not by reading the name. We introduce a grounding trial a shortcut cannot pass. We hold the pixels, the scene, and the robot pose fixed, change one word in the instruction, and require the arm to go to the newly named object. On this trial every frontier policy we tested selects the correct object 0–1% of the time, the axis the field collapses on. We trace the failure to imitation: the pretrained backbone resolves the named object in its language stream, but the action head routes position and never reads that binding. We cure it with a data recipe, not an architecture. We make the instruction the only predictor of the target and protect the pretrained grounding with a low-rank update; the same data under full fine-tuning destroys grounding (12.5%), and the low-rank cure installs it (94.5%). The cured policy, Galahad, obeys a renamed target on 200 of 200 trials and lands a median 2 mm from it, needs the pixels, and needs a real name. The released weight is one merged checkpoint that grounds seven referent-type axes at once (object identity, spatial, goal, color, category, negation, composition; obey rate 75–99%, occlusion $\leq$2% on every axis), and the recipe transfers to a second simulator, with the battery verified on a low-cost physical arm. And the principle unifies: on the object family, one set of weights trained with a prediction target carries two grounded outputs—its action *and* its prediction of the task-relevant future both obey the instruction, which no prior world-action model does. We release the model, the deconfounded datasets and their generator, and a one-command battery that checks any policy for the same shortcut. Every result in this paper was produced for less than $1,000 of cloud GPU rental; grounding, on this axis, is a matter of data, not scale.

1 Introduction

A vision-language-action (VLA) policy can pass a manipulation benchmark without reading its instruction (Figure 1). Replace the instruction with the string "xxx" and a frontier policy emits the same actions [Zhou et al., 2025]. It succeeds by moving to the object that usually occupies that position, not by resolving the words. The standard benchmarks reward this: they place the named object where the policy expects it, so a position shortcut and genuine language grounding earn the same score. On the one axis that separates them, object identity (rename the target, keep the scene), the published field selects the correct object 0–1% of the time.

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†Project page: `https://xn--7xa.monster/Galahad/`; code, models, and datasets: `https://github.com/phi-monster/Galahad` and `https://huggingface.co/phi-monster`.

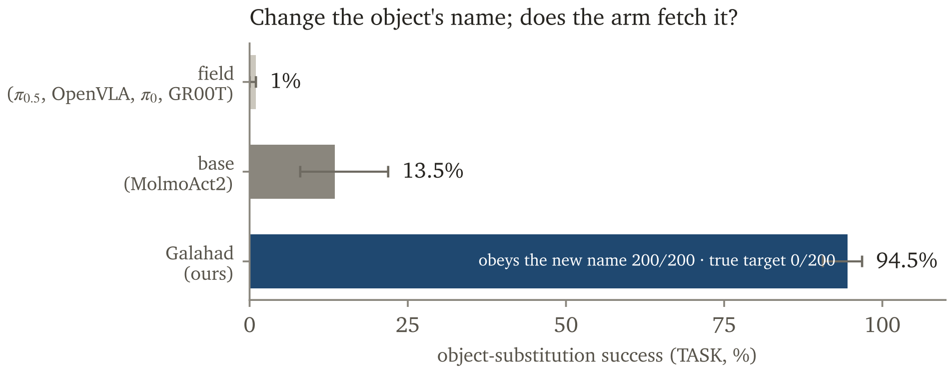

Figure 1: On the axis that separates grounding from a position shortcut, renaming the target object, the field scores 0–1%, the base model scores 13.5%, and Galahad scores 94.5%, obeying the new name on 200 of 200 trials and never fetching the object it was trained to fetch. Intervals are Wilson 95%.

We found this by trying to break it. We set out to build a cheat-proof adversarial standard for physical AI: a learned world-model attacker that would drive frontier policies to failure at their weakest point. The attacker could not beat a random poke. An adversary needs a victim with internal structure to exploit, and these policies expose none, because they do not read the instruction in the first place. The failure was the result: with no grounded policy to attack, there was nothing to measure. We stopped forging the sword and forged the knight.

Measuring grounding demands a trial a shortcut cannot pass, and neither standard signal is one. Task success is not: a policy that memorizes positions completes the task without reading the name. Vision-dependence is not: blanking the cameras collapses a position-memorizer as surely as a grounded policy. Our trial changes one word. Same pixels, same scene, same robot pose, we rename the target and record where the arm goes. A grounded policy moves to the newly named object; a shortcut moves to the same place regardless of the word. This one comparison, which we call SWAP-OBEY, converts a hollow "it failed" into a positive control ("it obeyed the new name"), and it is the instrument the rest of the paper builds on.

Grounding is curable, and the cure is a data recipe rather than an architecture. We make the instruction the only predictor of the target: every object appears as the target and as a distractor across randomized positions, so a position shortcut earns nothing and only reading the name pays. We protect the pretrained model's grounding with a low-rank update. Full fine-tuning on the identical data destroys grounding and returns 12.5%; a rank-32 adapter installs it and returns 94.5%. The frozen backbone already knows what the words mean. The recipe teaches the action to route what the backbone knows, and adds no coupling loss, no detector, and no architecture change.

This paper contributes six results, in the order of their weight.

1. **Galahad, a grounded policy.** On object substitution Galahad reaches 94.5% where its base scores 13.5% and the field scores 0–1%; it obeys a renamed target 200 of 200 times, landing a median 2 mm from it, and collapses when the cameras are blanked or the name is meaningless. The released weight is one merged checkpoint grounding seven referent-type axes (obey rate 75–99%, occlusion $\leq$2% on every axis), the recipe transfers to a second simulator, and the battery's controls verify on a low-cost physical arm.

2. **The trial.** A per-referent-type counterfactual protocol and an initial leaderboard on which the field scores 20–45%, a text-conditioned world model scores 64.7%, and Galahad scores near 100%. The protocol extends to the prediction side, a counterfactual that the world-model evaluation literature leaves open.

3. **The dataset and its generator.** An eight-type deconfounded design built from one principle and one generator per type, shipping the exam and the medicine in the same artifact.

4. **A diagnosis.** A black-box account of why imitation discards grounding, nailed three ways (literature, a circuit, and a null): the binding lives in the language stream and the action head routes position, and one word-direction injection flips the grounding.

5. **A causal cure and its boundary.** Low-rank protection and deconfounded data form a causal pair, a protection-capacity curve, and four boundary laws: an object's grounding costs only its appearance in a scene, the vocabulary is closed, grounding trades against shortcut robustness by design, and the cure requires a base whose backbone grounding is architecturally exposed to the action head.

6. **A conditioning gradient.** Text-conditioned generation grounds, action-conditioned world models do not, and imitation policies fail, one disease along one spectrum.

On the object family one set of weights carries both grounded outputs, its action and its prediction of the task-relevant future, which no prior world-action model does. We release the model, the deconfounded datasets and their generator, and a one-command battery that checks any policy for the same shortcut. The knight is the proof; the trial is what makes the proof un-fakeable.

2 A trial a shortcut cannot pass

A grounding claim is only as strong as the trial behind it, and two signals the field relies on do not constitute one. We state the trial before any result, because the trial is the contribution that makes every number un-fakeable.

2.1 Task success and vision-dependence do not measure grounding

A policy that memorizes where each object sits completes the task without reading the instruction, so task success alone certifies motor competence, not grounding. Blanking the cameras does not fix this: a position-memorizer collapses under blank cameras just as a grounded policy does, so an occlusion control (OCC) rules out a state leak but not a position prior. We show this directly in Section 3: the base model passes task success on the spatial suite at 99% and on the goal suite at 100%, and both scores are shortcuts. A grounding trial must vary the one thing a shortcut ignores, the word.

2.2 swap-obey: rename the target and watch the arm

Our primary instrument holds the pixels, the scene, and the robot pose fixed, changes one word in the instruction, and records which object the end-effector reaches. We draw the new name from an object that is actually present in the scene, so there is always a real target to go to, and we score two quantities: the success rate on the *original* target, which a grounded policy drives to zero, and OBEYED-NAME, the rate at which the end-effector lands on the *newly named* object.

OBEYED-NAME is the load-bearing readout. It converts a hollow "the policy failed" into a positive control, "the policy obeyed the new name," and only that direction distinguishes a policy that reads the instruction from one that rides the scene. We rotate the renamed decoy across every in-scene non-target so the trial cannot be passed by orienting toward one familiar object.

Two disciplines keep SWAP-OBEY honest. We verify the distance, not just the threshold: on the flagship, the end-effector lands a median 2 mm from the newly named object while sitting a median 9 cm from the originally trained one, and no trial falls in the 4–5 cm grazing band, so a 2 cm threshold gives the same verdict as a 5 cm one. We report a reach-probe on every failure, splitting it into REACHED-TARGET (the arm reached the right object and the grasp slipped, a motor failure) and WENT-DISTRACTOR (the arm went to the wrong object, a grounding failure), so we never record a motor limit as a grounding limit.

2.3 The battery asks three different questions

The three language faces are not redundant, because each removes a different way to cheat. OCC asks whether the policy uses vision (blank the cameras; success must collapse). SWAP-OBEY asks whether it follows the named object (rename the target; the arm must switch). NONSENSE asks whether it needs a name at all (replace the instruction with "xxx"; a policy that still succeeds is riding a scene prior, and its language is decorative). A wrong-pointer face (WOBJ) confirms the policy uses no injected coordinate. Success itself is lift-gated and init-false: the object must leave the table by a fixed margin, so a scene in which the target begins at the goal cannot score a do-nothing success. We measured the init value of every metric and required a do-nothing policy to score near zero before trusting any rate.

2.4 One protocol, every referent type

Object identity is the sharpest axis but not the only one, and the same counterfactual instantiates for each way an instruction can name a target: color, category, ordinal position, spatial relation, negation, and composition. In each case one word varies and the target must move to the object that word selects. This is the protocol behind the per-type leaderboard in Section 3 and the six-axis result in Section 6, and it is the exam half of the dataset artifact in Section 5.

2.5 The trial extends to the predicted future

A policy that also predicts the future exposes a second surface to the same test. We change one word and ask whether the predicted region of motion moves to the newly named object. This counterfactual, which we call C4-I, is the metric the world-model evaluation literature leaves open: existing leaderboards reward a prediction that *changes* under a new instruction (diversity), not one that changes *correctly* (control). Section 7 reports it for Galahad and Section 3 runs it across the generative world-model zoo.

3 No frontier policy passes the trial

We ran the trial across the field. The result is a leaderboard, not a wall: policies do not fail uniformly, they fail along a gradient set by how the instruction reaches the output.

3.1 The base has perfect motor and near-chance grounding

The base model, MolmoAct2 [Fang et al., 2026a], executes standard pick-and-place at 100% and selects the correct object under substitution 15% of the time. Measured cleanly at n=96, its object-substitution grounding is 13.5% $[8.0, 21.9]$. The gap between 100% motor and 13.5% grounding is the entire problem: the motor works, and the instruction does not drive it. This before number also settles the obvious question about Galahad, "how much was already there," at 13.5%.

3.2 The imitation field collapses on object identity

On the object-substitution axis the published imitation field scores 0–1%: $\pi_{0.5}$ at 1%, and OpenVLA, π_0, and GR00T at 0% [Zhou et al., 2025]. We do not rest the claim on one source. Yu et al. [2026] report frontier VLAs selecting the correct object at near-chance rates after controlling for grasp success; Chen et al. [2026] run thirty policies on an independent sim-and-real benchmark and find the best open-vocabulary instruction-following score is 1.67%; and our own search for a groundable victim, the search that started this project, returned none. Field-zero is triangulated, not asserted. The newest releases repeat the pattern: an open VLA reporting 98.7% on LIBERO reproduces its published 100% positive control on our harness, then drops to 10.0% $[4.3, 21.4]$ under object substitution—and completes the canonical task 52% of the time with the instruction replaced by "xxx" [Unitree Robotics, 2026]. The scene decides; the words are optional.

3.3 World-action models are a leaderboard, not a wall

We ran the battery on four open world-action models under a state-leak-clean protocol (blank cameras and blank proprioception both collapse success, so no model succeeds from a leaked state). Obey rates: InternVLA-A1.5 [Ma et al., 2026] 23.3%, UniVLA [Wang et al., 2026c] 20%, WorldVLA [Cen et al., 2025b] 25%, and RynnVLA-002 [Cen et al., 2025a] 45%, against Galahad near 100%. The best open world-action model reaches half of Galahad's grounding, and the rest sit near chance. None of their prediction heads can be verified natively: the only native action-conditioned frame head we could drive is instruction-blind by construction, which is itself the finding of the next subsection.

3.4 Grounding of the predicted future tracks the conditioning channel

The prediction side shows the same gradient (Figure 2), and it explains the world-action models' failure.

A world model whose future is conditioned on the *action* cannot read the instruction at the frame level: the name reaches the pixels only through a weak action head, and such models hallucinate a successful future even under a wrong action [Liu et al., 2026]. A world model whose future is conditioned on the *text* reads the name directly. We ran the C4-I counterfactual on a text-conditioned video world model [NVIDIA, 2025a] over held-out frames, hand-scored in pixels under a continuity protocol that rejects composited outcomes (the named object must leave its initial position along a gripper-continuous trajectory): it obeys the swapped object name 64.7% of the time $[41.3, 82.7]$ against a 25% chance rate, the strongest grounding in the world-model zoo. The tier replicates at the frontier: Cosmos 3's Nano model [NVIDIA, 2026], two generations newer with a stronger reasoner, lands at 77.8% $[61.9, 88.3]$ on the same protocol—the conditioning channel, not scale, sets the tier. The gradient is therefore one disease along one spectrum: text-conditioned generation grounds (64.7%), action-conditioned world models do not (frame head instruction-blind;

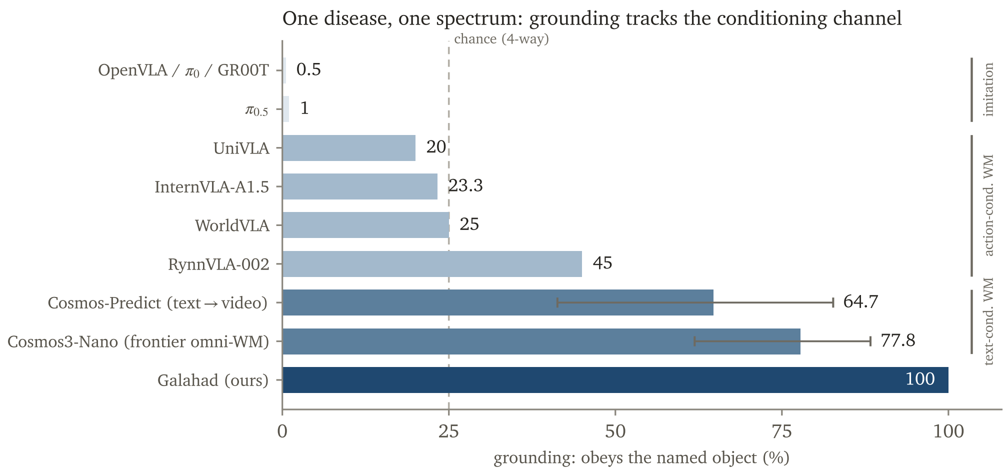

Figure 2: One disease along one spectrum. Grounding of the output tracks how the instruction reaches it: text-conditioned generation reads the name (Cosmos-Predict 64.7%), action-conditioned world models bottleneck it through a weak action head (20–45%), and imitation policies discard it (0–1%). Galahad grounds near 100%. Obey rate against a 25% chance rate.

action head 20–45%), and imitation policies fail (0–1%). Galahad sits above all of them on both faces.

3.5 A benchmark that rewards ignoring the instruction

One public robustness benchmark scores our cured model *below* its base, and the reason is a finding about the benchmark. On its seven perturbation factors the base scores 90.0% pooled and the cured model scores 76.4%, lower on every factor. The sharpest drop is the Language factor, 96.0% to 72.0%, a 24-point fall. That factor paraphrases the instruction and rewards a policy whose actions do not change, so a policy that ignores language is paraphrase-invariant and scores high. Our cure reads the instruction and therefore scores lower there. The Language axis measures language-*invariance*, not grounding, and the 24-point gap is the cleanest evidence that the two are opposite. We never report this benchmark's aggregate as our robustness; we report it as the measured price of reading the words.

4 Imitation discards the grounding it does not need

The cure is a data recipe, and it works for a reason we can point to in the network. Imitation learning on position-confounded demonstrations teaches the easy feature and starves the useful one, and the diagnosis is nailed three ways: the mechanism literature, a circuit we trace, and a null we report in Section 5.

4.1 The literature names the failure

Two lines predict exactly this failure. Causal confusion in imitation learning shows that behavior cloning latches onto non-causal shortcuts and that more information makes it worse, so the fix must be interventional [de Haan et al., 2019]. Gradient starvation shows that an easy, sufficient feature suppresses the gradient that would train a harder, more useful one, so once the position shortcut is

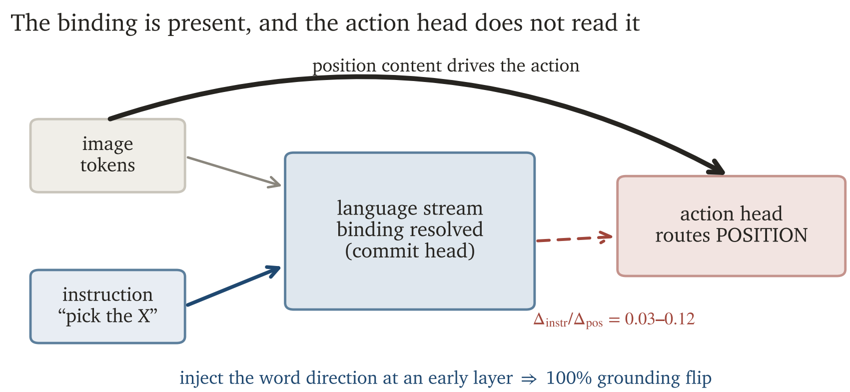

Figure 3: The lesion imitation leaves. A strong backbone resolves the named object in its language stream, but the action head routes the position content and leaves the language binding unread: patching the instruction content moves the action 0.03–0.12 as much as patching the position content. One word-direction injection at an early layer flips the grounding entirely.

sufficient the model provably stops learning the language path [Pezeshki et al., 2021]. Independent measurements confirm the endpoint. Fang et al. [2026c] show frontier VLAs act from vision without the instruction: given vision alone, with the instruction removed, OpenVLA-OFT still scores 69% and $\pi_{0.5}$ 52%, while given the instruction alone, with vision removed, both collapse below 1%. Lian et al. [2026] reach the same conclusion in an information-theoretic frame: when the instruction is predictable from vision, the conditional information the action carries about it vanishes and the policy degenerates to a vision-only prior. The action does not read the instruction.

4.2 The binding is present, and the action head does not route it

The grounding is not missing from the model; it is missing from the action (Figure 3). In a strong backbone the named object is resolved in the language stream and committed by a single late head, while the vision tokens remain instruction-blind. We test whether the action head uses that binding by patching the language-stream residual from one instruction into another forward pass and reading the change in the action. The position content moves the action a lot and the instruction content moves it almost nothing: across eight cloned policies the ratio of instruction-driven to position-driven action change is 0.03–0.12. The action head routes position and leaves the language binding on the table. This is the lesion the cure repairs, and it is why full fine-tuning, which reshapes the action head on confounded data, destroys grounding rather than installing it.

4.3 One direction flips the grounding

The binding is causal, not decorative. Injecting a single color-word direction at an early layer flips which object the model grounds to 100% of the time, while injecting the mean read-out direction at the commit layer flips nothing, because the read-out encodes scene-specific position and the identity lives at the word. This extends activation-level steering of VLAs [Häon et al., 2025] from motion to object identity. Grounding is a controllable computation in the language stream that the action, as trained by imitation, declines to read.

The cure closes the lesion. The open-loop instrument that exposes the lesion reads it closed after the cure. On the base, the action aligns with a named object when it is the scene's canonical target (mean cosine +0.83) and poorly when the same object is a distractor (+0.47, with 22% of distractor instructions pointing *away* from the named object); on the cured weight the role gap vanishes (+0.92 against +0.94, nothing pointing away). The action reads the name regardless of the role the training scenes assigned the object.

Scope. We measured the circuit and the patching ratio on SmolVLA and Qwen2.5-VL, so this section explains why the field fails and why a data intervention is the right axis, and does not claim to trace Galahad's own internal head-by-head circuit. The behavioral evidence in Sections 3 and 6 carries the claims about Galahad.

5 A data recipe installs grounding; no training-time surrogate does

The cure changes the data and protects the model, and nothing else. We show the two ingredients are a causal pair, and we show that four alternatives a reader would reach for first all fail.

5.1 The deconfounding principle: the instruction is the only predictor

We build demonstrations in which the instruction is the only variable that predicts the target. Every object appears as the target and as a distractor across randomized positions, and the distractors are always in frame. A position prior earns nothing, because the target is not where it was last time; a scene prior earns nothing, because every scene contains every object; only reading the name pays. The principle is one sentence, and it is the whole method. Applying it turns the base's 13.5% into a grounded policy.

5.2 Low-rank protection is structural, not a small step

The pretrained backbone already resolves the words (Section 4), so the training must install the routing without overwriting the resolution. A low-rank update [Hu et al., 2022] does this and full fine-tuning does not: on the identical data, full fine-tuning returns 12.5% (grounding destroyed) and a rank-32 adapter returns 94.5%. The protection is structural, a property of the rank, not of the step size: doubling the learning rate barely moves the result (68.75% to 66.7% on the development data), while releasing the rank breaks it. The protection-capacity curve rises then plateaus, 56.7 / 65.0 / 68.75 / 68.3 at ranks 8 / 16 / 32 / 64 on the development data, so rank-32 is a measured choice with no headroom above it; on clean data the curve is flatter and higher (85 at rank 8, 93 at rank 32). Low rank preserves the grounding; the deconfounded data trains the action to read it. Neither ingredient works alone.

5.3 Four alternatives fail, and the failures are the argument

A grounding objective, a training-time regularizer, and more data are the obvious substitutes, and each one is dead (Figure 4).

Coupling supervision harms monotonically. Supervising the action head's attention to peak on the named object [Jia et al., 2026], the mechanism a reader would expect to help, degrades grounding in a clean dose-response: 68.75% with no coupling, 35% at weight 0.2, 21.7% at weight

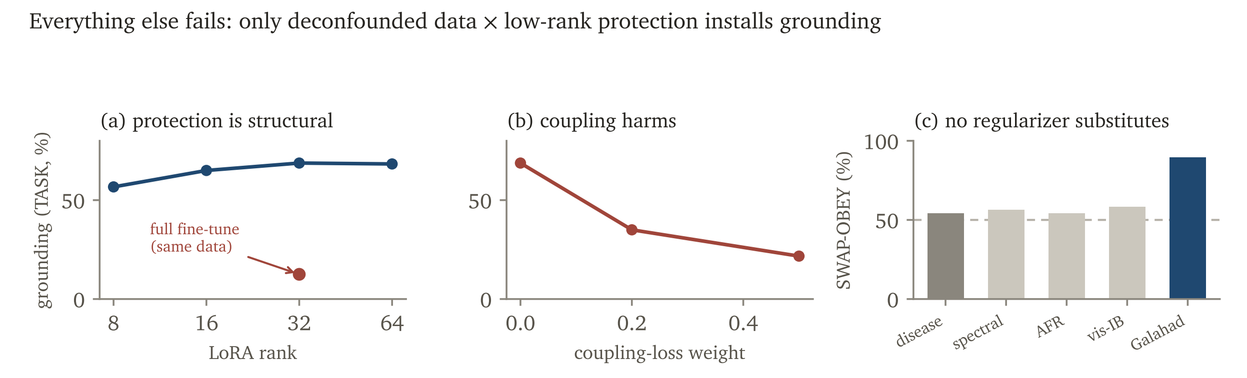

Figure 4: Only deconfounded data with low-rank protection installs grounding. (a) Protection is a property of the rank: the capacity curve rises then plateaus, and full fine-tuning on the same data collapses to 12.5%. (b) Coupling supervision harms in a dose-response. (c) Three training-time regularizers on confounded data all sit at the disease level; none approaches the deconfounded recipe.

0.5. The frozen backbone already knows where to look; forcing the action expert to be told over-constrains its read-out and breaks it.

No training-time regularizer substitutes for the data. We trained three regularizers on the original confounded data, the direct test of whether the paradigm can be fixed without the anti-shortcut dataset: a spectral-decoupling penalty on the action, an action-head loss reweighting, and a one-sided visual information bottleneck. Every arm at every hyperparameter lands at the confounded-data disease level (task 2–25%, obey 48–62%) and none approaches the deconfounded recipe (task 60%, obey 90%). A strong penalty raises a metric only by damping the motor. The regularizers cannot manufacture grounding from confounded data; the deconfounded data can.

Data scale alone is dead. On the minimal testbed, doubling deconfounded demonstrations moves object selection from 52% to 50.5%. Volume without the routing fix raises the blind floor, not the grounding.

5.4 One principle, one generator per referent type

The principle is universal and the design is per-type: color needs color orthogonal to position and shape, spatial needs side orthogonal to identity, and negation, relation, ordinal, and composition each break a different confound. All share one scripted-oracle generator and one battery, so each type is a small design, not a new method. The generator ships with the paper, and it produces the exam (the counterfactual battery) and the medicine (the training set) from the same code.

6 Galahad grounds language un-fakeably

We evaluate Galahad on the trial of Section 2. It passes on object identity at a rate no prior policy approaches, its released weight grounds seven referent-type axes in one merged checkpoint, and its boundaries are measured laws.

Table 1: Main battery, object substitution, all ten slots, n=200, seed 1000, on the object-family specialist weight. Galahad passes every face; the base has the motor and lacks the grounding.

face	question	Galahad	base
TASK	fetches the named object?	**94.5%** [90.6,96.8]	13.5% [8.0,21.9]
SWAP-OBEY	goes to the newly named object?	**200/200**	n/a
SWAP true-tgt	avoids the trained object?	**0/200**	n/a
OCC	needs vision?	0/20	0/20
NONSENSE	needs a real name?	4.5%	n/a
motor	vanilla pick-place intact?	n/a	100%

6.1 The main battery

Table 1 is the flagship result on the object-substitution axis, all ten slots, n=200. Galahad succeeds 94.5% of the time where its base scores 13.5% and the field scores 0–1%. It obeys a renamed target on 200 of 200 trials and never completes the originally trained pick (0 of 200). It needs the pixels (blank cameras collapse it) and it needs a real name (the string "xxx" drops it to 4.5%). The 4.5% is a reported residual, not a suspicious zero: one scene retains a scene-to-target prior strong enough to act without a name, and we leave it in the number.

Takeaway. The base carries the motor and lacks the grounding; the recipe installs the grounding and keeps the motor. The advance is 13.5% to 94.5% with the pick-and-place unchanged, which is why the gain is grounding and not a stronger policy.

6.2 The position-prior story is dead, and the one cliff is a motor wall

The benchmark's own perturbation suite refutes the "it memorized where things sit" reading. Displacing the objects 21 cm costs nothing (96.7%), so the policy tracks objects visually rather than by remembered location. At a 35 cm displacement the score craters to 6.7%, but the reach-probe shows the arm still reaches the correct object every time (REACHED-TARGET 3/3) and doubling the step budget does not rescue it. The grounding survives the displacement; the low-level grasp fails at the edge of the workspace. The cliff is a motor limit of the base, not a grounding limit of the recipe.

6.3 Three axes, three shortcuts, all cured

Object identity is one of three shortcuts the base hides, and the trial exposes each one. Task success plus occlusion pass on all three while SWAP-OBEY fails on all three, which is the point of Section 2: only the swap distinguishes them. On the spatial suite the base scores 99% on task and 50% on SWAP-OBEY, exactly chance for two bowls, so the 99% is a canonical-bowl shortcut; the cure raises SWAP-OBEY to 86% at a motor cost (task 51%, grasp-limited, occlusion-clean). On the goal suite the base scores 100% on the canonical scene but collapses to 31.2% on held-out scenes with the objects displaced and the instruction held fixed, reaching for the remembered position; the cure raises the held-out rate to 79.2% [65.7, 88.3] with occlusion at 0/48 and the reach-probe going to the moved object 12/12. All three axes are base shortcuts, none pre-grounded, all cured by the same recipe.

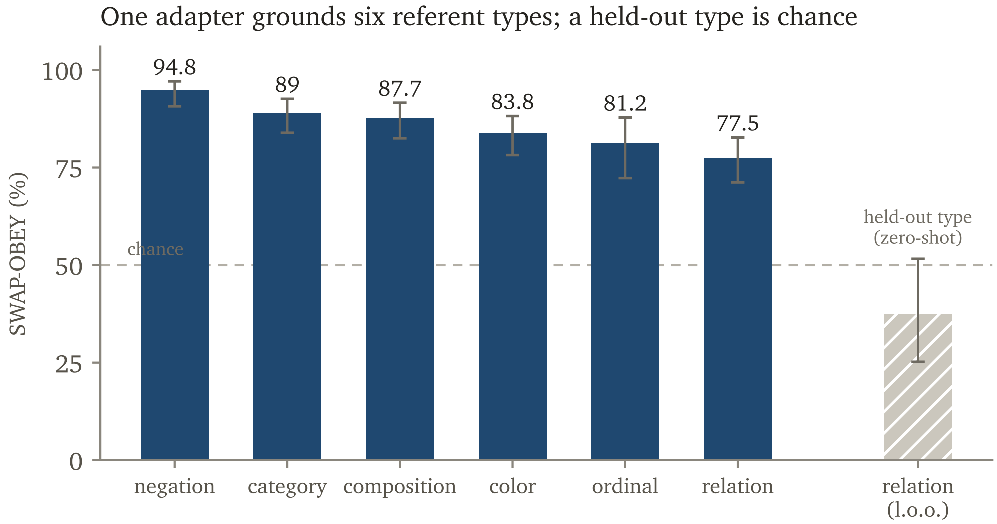

Figure 5: One adapter grounds six referent types at once (SWAP-OBEY 77–95%, occlusion 0 on every axis), and generality stops at its edge: a type held out of training grounds at chance. Intervals are Wilson 95%; the dashed line is chance.

Table 2: One adapter, six referent-type axes, $n \geq 96$ per axis. SWAP-OBEY is grounding; task is grounding times motor. Leave-one-out: a type held out of training grounds at chance.

axis	category	color	ordinal	relation	negation	composition
SWAP-OBEY	89.0	83.8	81.2	77.5	**94.8**	87.7
OCC	0	0	0	0	0	0

6.4 The recipe transfers to a second simulator

The recipe is not tuned to one environment. On a second simulator with a different renderer and object set, the same recipe grounds the object name at SWAP-OBEY 44–60% across two seeds against a 20% chance rate, with occlusion at 0. The task rate is motor-limited at 14–28%, the same "grounding works, motor is the ceiling" pattern as the spatial and goal suites, and we cite the swap range rather than a single point because the cross-simulator motor is seed-sensitive.

6.5 One adapter grounds six referent types

Table 2 and Figure 5 are the joint-training generality result: a single adapter, trained across six referent-type axes at once, including two (ordinal, relation) beyond the release set, grounds all six.

Every axis clears SWAP-OBEY 77–95% with occlusion at 0, so one model reads color, category, ordinal position, spatial relation, negation, and composition. The task rates (26–51%) are the motor ceiling, and SWAP-OBEY is the pure grounding metric.

The boundary law. Generality holds within trained coverage and stops at its edge. A model trained on five types and tested zero-shot on the sixth grounds it 37.5% of the time, at chance, against 75% when that type is in training (disjoint intervals). The recipe grounds the axes it has seen and does not transfer to a fully unseen axis. We write "general within trained coverage"

Table 3: The release: seven axes measured on the shipped merged checkpoint. SWAP-OBEY is grounding; task is grounding times motor and is bench-dependent. Spatial's obey count is backed by a 6.2× distance separation (d_{named} median 20 mm vs d_{true} 124 mm), required because its two referents are identical bowls.

axis	SWAP-OBEY	OCC	task
object	**99.0%** (198/200; true-target 0/200)	0%	80.0%
spatial	**obeyed 187 / went-true 0**	0%	55.0%
goal (displaced referent)	**83.3%** (base 31.2%)	0%	—
colour	**88.9%**	0%	22.2%
negation	**89.6%**	0%	16.7%
category	**78.0%**	2%	20.0%
composition	**75.0%** (colour 87.5 / size 62.5)	0%	2.1%

and never "zero-shot to any new axis." The edge is the type, not the composition: the same adapter zero-shots a never-trained *composition* of two trained types—"the second red one from the left"—at 64.2% where its adapter-ablated base scores 16.1% (paired +55.6 points, McNemar $p \approx 10^{-5}$, paraphrase-controlled). Its occlusion control collapses only partially (36%), the positional prior ordinal phrasing admits, so the evidence is the paired base gap. Trained types compose; an untrained type does not transfer.

6.6 The word selects the motor program

An eighth referent type grounds the verb. On a deconfounded lift-versus-slide design, the same object under both verbs, the cured policy's verb selectivity is +44.3 points: told to lift, it lifts 46.3% of the time; told to slide, it lifts 2.0% (disjoint intervals), and the base's selectivity is +0.0. The lift program collapses to zero with the cameras blanked, and the render shows a real grasp-then-raise against a real push-along-surface. The magnitude is motor-limited (pooled obey 37%, a grasp-stability wall), and the claim is the selectivity: the word chooses which motor program runs. Joint training dilutes this axis (+6 points in the unified weight), so the released checkpoint does not carry it and we report the single-axis result.

6.7 The release grounds seven axes in one weight

The released artifact folds seven referent-type axes into one merged checkpoint, and Table 3 is measured on that shipped weight, not on its adapters. Every axis grounds (SWAP-OBEY 75–99%) and every axis is vision-gated (occlusion $\leq$2%; the zeros are live measurements, since the same scorers read non-zero under other conditions in the same runs). On object identity the release obeys the renamed target 198 of 200 times and never fetches the trained object, matching its adapter (98.3%), while completion reads 80.0% against the specialist's 94.5%: the reach-to-completion gap is motor (reach-to-true-target 90.5%), the measured price of carrying seven axes in one set of weights, and grounding pays none of it.

The task column is bench-dependent, in both directions. The low task rates on the second simulator's axes are the bench, not a regression: the published single-axis anchors on the same bench read task 26–51% against SWAP-OBEY 77–95%, and the release sits 10–15 points under those task anchors, the same unification price as the object axis, while its grounding sits at or above them; a RoboCasa axis is never reported on task alone. Spatial is never reported on task alone either, in the opposite direction: the uncured base scores 99% on task and exactly chance on SWAP-OBEY, so task alone inverts the conclusion, and the release is the only one of the two reading the

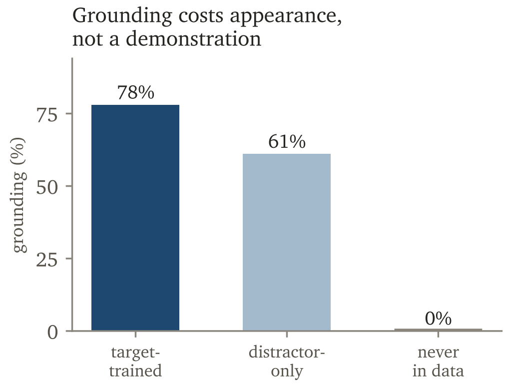

Figure 6: The scaling economics of the recipe: an object's grounding costs its appearance in a scene, not a grasp demonstration. Target-trained 78%, distractor-only 61%, never-in-data 0%.

spatial phrase. And composition decomposes: the colour attribute follows at 87.5%, matching the single-attribute axis, while size follows at 62.5%, so the short leg is size, which is actionable where a pooled 75% is not.

6.8 The boundaries are laws

Three boundaries are findings.

An object's grounding costs its appearance, not a demonstration. Grounding follows an exposure gradient (Figure 6): an object trained as a target grounds at 78%, an object seen only as a distractor grounds at 61%, and an object never in the data grounds at 0%.

The marginal cost of a new object's grounding is that it appear in a scene, not that it be grasped, which is the scaling economics of the recipe.

The vocabulary is closed. An object removed from training entirely, target and distractor, grounds at 0 of 16; told its name, the arm goes to a trained object instead, while trained-object controls hold at 75%. The recipe grounds a closed vocabulary and does not extend to a never-seen word. Open-vocabulary grounding is a different problem we do not claim.

Grounding trades against shortcut robustness by design, quantified in Section 3: the cured model scores below its base on a perturbation benchmark whose Language axis rewards ignoring the instruction. The trade is the measured price of reading the words, not a regression.

7 One weight, two grounded faces

A world-action model is meant to do two things at once: act, and predict the world it acts in. The class exists [Cen et al., 2025b,a, Wang et al., 2026c, Ma et al., 2026, Li et al., 2025c, Zhu et al., 2025], but no member of it has been shown to ground language on either face, let alone both; our battery puts the best of them at 20–45% (Section 3). Galahad's object-family weight is one set of weights whose action *and* whose prediction of the task-relevant future both obey the instruction. That is the first verifiably grounded unified model, and this section is its evidence; the seven-axis release weight of Section 6 carries the action face only, and every claim in this paper is pinned to the weight it was measured on. We pre-delimit "unified" as a single checkpoint with two grounded outputs, not a policy with a bolt-on latent predictor [Assran et al., 2025] and not a multi-model video platform [Liao et al., 2025].

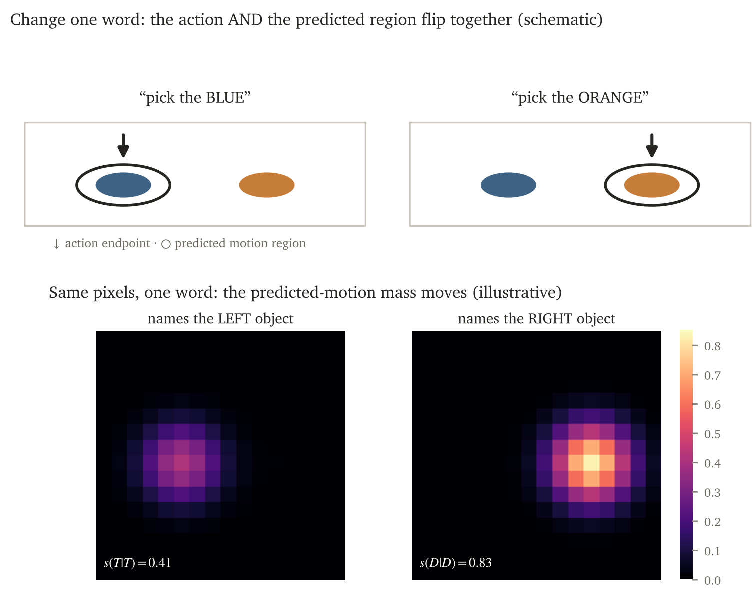

Figure 7: Two faces of one weight. Top: changing one word moves both the action target and the predicted region of motion to the newly named object (schematic). Bottom: the predicted motion mass over two objects, same pixels, one word changed; the mass moves two orders of magnitude (illustrative of the measured $s(T|T)$=0.41, $s(D|D)$=0.83).

7.1 One set of weights, from the released recipe

The unified model is one set of weights, not a cascade: the low-rank update folds into the base and the foresight head lives in the same checkpoint, the fold changing the action by a bounded 1.17×10^{-2}. A chained demonstration on this weight moves both faces in one forward pass: changing one word flips the predicted region of motion *and* the action endpoint to the newly named object, the end-effector landing 0.3–0.5 cm from it; the demonstration's frames and episodes are public. The checkpoint we release for this weight is a rebuild from the released generator, data, and training code, and it reproduces the battery at verification n (task 95%, SWAP-OBEY 38/40, true-target 0/40, occlusion 0/40) — the experiment is reproducible end-to-end, and the rebuild is the proof. The action-only variant is an ablation row rather than a second product.

7.2 The prediction face reads the instruction

Table 4 and Figure 7 are the center result on the prediction side. Holding the pixels fixed and changing one word, the predicted region of motion moves to the newly named object 58.3% of the time against a 25% chance rate ($p < 10^{-6}$), and the predicted mass over the two objects moves by two orders of magnitude. Blanking the cameras kills the effect (0 of 48), so the prediction reads the pixels rather than looking up a name-to-position table. The prediction is a grounded predictor of which object will move, which the world-model literature's own functional, intervention-aware definition counts as a world model [Wang et al., 2026a]. It is not action-conditioned and not rollable, so it is not a runtime imagination and we do not present it as one; foresight branches of this kind are routinely removable at inference [Fang et al., 2026b], and the value we claim is a verified grounded prediction, not a simulator.

Table 4: The prediction face, object-family weight, n=48. The predicted future obeys the instruction and needs the pixels. The action face is more grounded than the prediction face, and we report both.

counterfactual	result	chance
C4-I switch (one word, region moves to named object)	58.3% ($p < 10^{-6}$)	25%
C4-III blind switch (same, cameras blank)	0/48	25%
action face SWAP-OBEY (Section 6)	100%	n/a

7.3 The two faces are not equally grounded

The action face obeys at 100% and the prediction face switches at 58.3%. Both clear chance by a wide margin; they are not equal, and the asymmetry is the result—neither face borrows the other's number. The prediction's failures concentrate in the box-shaped look-alikes that also strain the action face. The headline is two grounded faces of unequal strength.

7.4 Unification costs no action accuracy

Adding the prediction face does not tax the action. Trained jointly from the start, the two-branch model ties the action-only control exactly (93.3% each, gap 0.0 points) with the foresight head genuinely learning (region overlap 0.47–0.71 against a 0.05 floor). Because that bench is saturated, we re-ran the comparison on a headroom bench where a tax could show: across action-only, two-branch, and read-back-masked variants the task rates are 48 / 53 / 47% and the grounding rates are 87 / 88 / 86%, intervals fully overlapping. The unification neither dilutes grounding nor costs task success, and the comparison is bounded by seed variance rather than by any between-arm effect: a single arm swings 66.7–90.0% across two seeds on the confusable slot, wider than any gap between arms. We report this as a clean null: the two-branch model does not beat the control either, and the spine of the paper is that both faces are grounded, not that one helps the other.

8 The battery transfers to a physical arm

We verify the battery on deliberately modest hardware: an SO-101 arm, a low-cost 3D-printed platform, fine-tuned with the same recipe on deconfounded real demonstrations. The controls transfer — blanking the cameras collapses the policy to a single instruction-independent attractor, while with vision the reach direction follows the instructed color — and successes are adjudicated by pixel co-location of gripper and target in the side camera. Grounding and actuation dissociate exactly as this platform predicts: reach-obey 10/21 (48% $[28, 68]$, chance 33%), grasp-obey 5/15; the grasp is what the hardware breaks. Footage is on the project page.

9 Related work

Vision-language-action policies and their benchmarks. A generation of open VLAs maps pixels and instructions to actions [Brohan et al., 2023, Kim et al., 2024, Black et al., 2024, Physical Intelligence, 2025, 2026, Shukor et al., 2025, NVIDIA, 2025b, Lee et al., 2025, Fang et al., 2026a], trained and scored on standard manipulation suites [Liu et al., 2023, Li et al., 2024, Nasiriany et al., 2024, Tao et al., 2024, Chen et al., 2025] in a common toolchain [Cadene et al., 2024, Chi et al., 2023]. These suites measure motor competence: they place the named object where the policy expects it, so a position shortcut and grounding earn the same number. Galahad is a low-rank fine-tune of one such base [Fang et al., 2026a]; the contribution is the recipe, the ruler, and the finding, not the backbone.

Grounding evaluations and shortcut learning. Perturbation and paraphrase benchmarks expose the gap our trial targets: moving or renaming the object collapses the field to near zero on object identity [Zhou et al., 2025, Fei et al., 2025, Kim et al., 2026, Wang et al., 2026b, Emukpere et al., 2026], controlling for grasp shows target selection at chance [Yu et al., 2026], a thirty-policy sim-and-real benchmark reports best open-vocabulary instruction following at 1.67% [Chen et al., 2026], broader robustness suites report the same fragility [Pumacay et al., 2024, Wang et al., 2025], and occlusion studies confirm that fixing perception does not fix grounding [Li et al., 2026]. Concurrent work diagnoses the vision shortcut and adds an inference-time guidance fix [Fang et al., 2026c]. We subsume this line on three axes: a positive control (the arm must go to the newly named object, not merely score a drop), a data recipe that reaches 94.5% where inference-time guidance tops out far lower, and coverage across seven referent-type axes in one released weight, two simulators, and a physical arm.

Why imitation discards grounding. Causal confusion and simplicity/gradient-starvation bias predict that behavior cloning latches onto a sufficient easy feature and starves a harder useful one [de Haan et al., 2019, Wen et al., 2020, Pezeshki et al., 2021, Shah et al., 2020, Geirhos et al., 2020], a pattern now measured directly in robot policies [Fang et al., 2026c, Lian et al., 2026, Xing et al., 2025]. We localize it: the object-name binding lives in the language stream while the action head routes position, and one direction flips it, extending activation-level steering of VLAs [Häon et al., 2025] from motion to object identity.

Injecting grounding, and why we do not. A parallel line adds grounding through the architecture: an explicit point or box injected into the action head [Tsai et al., 2026, Yu et al., 2025], an attention-alignment object head [Jia et al., 2026], or a reasoning branch [Sun et al., 2026]. We trained these on our data and cut them, and install grounding with data instead: counterfactual, role-balanced demonstrations [Pitis et al., 2020, Ameperosa et al., 2025] under a low-rank update [Hu et al., 2022] that protects the pretrained model, the gradient-isolation principle of Driess et al. [2025] by a cheaper means.

Reinforcement learning is not the fix. RL sharpens what a base already samples but does not add a capability it lacks [Yue et al., 2025, Liu et al., 2025]; the apparent RL-over-SFT generalization gap is largely a data artifact that referent diversity closes [Chu et al., 2025, Lin et al., 2025]. Grounding is missing from the base's action, so we supply it with data, not reward.

World-action and world models. Unified models that predict a future alongside actions are an established class [Cen et al., 2025b,a, Wang et al., 2026c, Ma et al., 2026, Li et al., 2025c, Zhu et al., 2025], so we claim the first *verifiably grounded* such model and pre-delimit unified as single weights with two grounded outputs: a bolt-on latent predictor [Assran et al., 2025] or a multi-model video platform [Liao et al., 2025] is neither. Across this zoo the predicted future does not obey the instruction, because action-conditioned foresight is a training-time regularizer, removable at inference [Fang et al., 2026b], that hallucinates success under a wrong action [Liu et al., 2026]; whether such models even generalize better than VLAs is contested [Zhang et al., 2026]. We adopt the functional, intervention-aware definition of a world model [Wang et al., 2026a, Hansen et al., 2024]; world-model leaderboards reward perceptual quality over instruction-correctness [Shang et al., 2026, Li et al., 2025a, Duan et al., 2025, Hu et al., 2025, NVIDIA, 2025c,a] — the counterfactual-controllability whitespace our prediction-face metric fills.

A data recipe on a base. Galahad follows the template of instruction-following-from-a-base-plus-a- data-recipe [Taori et al., 2023, Wang et al., 2023]: we take the strongest open base and add the deconfounding principle, the un-fakeable ruler, and the grounded model itself.

Adversarial policies. This project began in the adversarial-policy tradition [Gleave et al., 2020]: a learned attacker exploits a fixed victim. A policy that does not read its instruction is a hollow victim with no structure to attack, so the grounded model revives the embodied contest [Li et al., 2025b] that a hollow victim makes degenerate.

10 What a data cure can and cannot do

The result is a policy that grounds language by data, and its edges are as informative as its center. We map them.

Data is the source of grounding, and a training-time regularizer is not. We tested whether a penalty, a reweighting, or a bottleneck could force grounding on confounded data, and all three failed (Section 5). Grounding came only from making the instruction the sole predictor of the target. The implication is precise: the deconfounded data is the active ingredient, and the low-rank update is the protection that lets the pretrained resolution survive.

The boundaries are the scaling economics. Grounding costs an object's appearance, not a demonstration (the exposure gradient), which is what makes broadening coverage cheap. The vocabulary is closed, so coverage must be grown rather than assumed. Generality holds within trained coverage and stops at an unseen referent type. And the coverage itself is hand-built: every referent type in this paper was deconfounded by a person who had anticipated its shortcut—one design per confound, colour against position and shape, ordinals against absolute position, negation against keyword matching. Together these say the recipe scales by coverage at low marginal cost, does not leap to categories nobody designed for, and keeps a designer in the loop. That is a map, not an apology—and the map points at the successor.

The cure needs a base that can route. The same recipe on a second base is a null, and the null has a mechanism. On $\pi_{0.5}$, the low-rank cure under both adapter placements—action expert only, and VLM plus expert—leaves SWAP-OBEY at chance (53.3% against a 46.7% base, n=60 each); on a frontier omnimodal world model's action head the cure is a null as well (a pilot). The shared factor is architectural: both condition their action expert on a pooled summary of the backbone, so the token-level binding the recipe protects is never exposed to the action head, and there is nothing to route. MolmoAct2's per-layer interleaving exposes it, which is why the cure lands there; an open VLA whose flow head cross-attends the full token sequence sits between the two poles, and we predict it curable. The recipe is a data cure conditioned on a routable architecture, not a universal patch—and the boundary sharpens the diagnosis: the disease was never missing knowledge, it is a severed route.

Grounding is one kind of robustness, and we claim one kind. Reading the instruction gives robustness to renaming, paraphrase, and displacement, because those are the semantic axis. It does not give robustness to camera, lighting, and dynamics, which are orthogonal, and the cured model trades shortcut robustness for grounding on a perturbation benchmark by design. We claim the semantic axis and earn the perceptual axis later through domain randomization, never by asserting that grounding implies robustness.

The victim the attacker needed. This project began as an attack that found no target: a learned adversary could not break policies that do not read their instruction. A grounded policy is a victim with internal structure to exploit, so Galahad reopens the embodied adversarial contest that a hollow victim made degenerate. The grounded model is not the end of the attack line, it is the substrate that makes the attack line non-trivial.

One principle, two grounded outputs. The conditioning gradient of Section 3 is a single spectrum: generation conditioned on text grounds the instruction, world models conditioned on action do not, and imitation policies fail. Galahad grounds both its action and a structured prediction of the future by applying the deconfounding principle to the prediction target, which is why one set of weights obeys the instruction on both faces.

Where we are going. The evidence in this paper places the disease outside every model class. The backbone grounds the referent (Section 4); generation conditioned on text grounds it (Section 3); the failure appears wherever an action objective can satisfy itself without reading the instruction—in imitation policies and in action-conditioned world models alike. The disease is not a property of vision-language-action models and not a property of world models: it lives where language enters an action objective. The conclusion we draw is not to couple these models more tightly but to separate them completely: grounding belongs to general vision-language and world models, which this paper shows already possess it, and action belongs to a model that never reads language at all. An action model with no linguistic input has nothing for causal confusion to discard—the shortcut this paper documents cannot form, by construction rather than by data design, and no referent type ever needs to be hand-designed again. Separation also removes the constraint that has capped robot-data scale, which was never pixels but aligned language: instructions across datasets are templated, mismatched, or absent, so the data cannot be pooled—while a language-free action model can train on every robot episode ever collected. We will report that work separately; this paper is the account of the disease, the ruler, and the first policy to survive the trial.

11 Release

We ship four artifacts, each usable on its own. The **model** ships as two checkpoints: the seven-axis merged release weight, and the object-family dual-output weight of Section 7, rebuilt from the released recipe and verified against its battery. The **datasets** are the deconfounded referent-type sets in a standard format, the first large deconfounded manipulation data, generated in simulation and renewable. The **generator** produces the exam and the medicine from one principle, so a new referent type is a small design rather than a new pipeline. The **battery** is a one-command grounding check, a lie-detector that any policy can be run through: it reports SWAP-OBEY, OCC, NONSENSE, and the reach-probe, covering both the action face and the prediction face that the field currently leaves unmeasured. Everything is open, permissively licensed, and reproducible from one command, so the ruler is adopted by being run, not by topping a leaderboard.

12 Conclusion

We set out to attack physical AI and found nothing grounded to attack, so we built the grounded policy the field was missing. A trial that a shortcut cannot pass shows that frontier policies read position rather than the instruction, a data recipe that makes the instruction the only predictor of the target cures it, and the cured policy grounds seven referent-type axes in one released set of weights, across two simulators, and on the object family both its action and its prediction of the future. The knight sits in the seat; the trial is what proves it is the knight and not another pretender.

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A The measurement discipline

The word cheat-proof in the title is a claim about our own process as much as about the models, and this appendix is the evidence. Every headline in this paper survived an attempt to disconfirm it, and four times the attempt succeeded and changed the number. We record the four, because a benchmark is only as trustworthy as its authors' willingness to catch themselves.

A too-clean number is a smoke-bomb until a disconfirming control passes. Our standing rule is that a 100% or a 0% is a suspect, not a trophy, until a control rules out the instrument. We verify in the modality the truth lives in: a scalar can describe a broken world, so we render the trained-policy rollout and look at the pixels rather than trusting the number. The four cases below are this rule paying out.

Case 1: the probe that verified grounding was ill-posed. An early instruction-swap probe fed each task the next task's instruction verbatim, and on most slots the named object was not in the scene, so a zero was uninterpretable in both directions. We rebuilt it as SWAP-OBEY: the decoy is drawn from the objects actually present, and the read-out scores whether the arm went to the named decoy. Every grounding number from the ill-posed probe was void, and the well-posed probe is the one this paper uses.

Case 2: a structural zero in the data masqueraded as a policy limit. Two slots scored zero for weeks and were filed as a round-bottle motor limit. The reach-probe said the arm reached the target and failed to grasp, so we called it motor. A two-episode trace of the demonstration generator found the real cause: the scripted oracle rammed the tall bottle at hover height, knocked it over, and closed on air, so those slots had no demonstrations to learn from. The fix was in the generator, and the headline moved 22 points with no change to the method. A structural zero in the training data is not evidence about the policy.

Case 3: our own reported effect did not replicate. We reported that the two-branch model taxed grounding by 21.6 points, with a clean mechanism and a per-object reach-probe to match. A second seed erased it: the tax lived inside single-seed variance on one look-alike slot, and its sign flipped across seeds. We retract the tax and report the null (Section 7), and we flag single-seed pilots as pilots.

Case 4: a metric can lie by deflating, too. A color axis read task 0% and looked like an honest motor failure. It was a scorer aggregation bug; the corrected rate is 37.5%, and the grounding was intact the whole time (SWAP-OBEY 83.8%). We now verify a suspiciously low number as hard as a suspiciously high one.

Environment-level smoke-bombs. The success predicate itself can be gamed by the environment: an object that spawns inside the goal region scores a do-nothing success, and blank cameras then fail to collapse it. We measured the init value of every metric and lift-gated success (the object must leave the table), so a do-nothing policy scores near zero by construction.

B Referent-type generator designs

One principle (the instruction is the only predictor of the target) instantiates as one small design per type, sharing a scripted-oracle generator and the battery. Table 5 lists the confound each type must break.

Table 5: One generator, one design per referent type: the confound each breaks and the counterfactual that verifies it.

type	confound broken	swap counterfactual
object identity	position, scene	rename the target object
color	color $\perp$ position $\perp$ shape	change the color word
category	category $\perp$ instance $\perp$ co-appearance	change the category word
ordinal	ordinal $\perp$ absolute position $\perp$ identity	change the ordinal word
spatial relation	side $\perp$ identity	name the other referent
negation	keyword-match	flip the negated attribute
composition	single-attribute lookup	change one attribute in the pair
goal	remembered destination	displace objects, hold the goal word

C Compute

All models in this paper train and run on commodity GPUs, and every result was produced for less than $1,000 of cloud GPU rental. Each Galahad variant trains with 8-way data parallelism on NVIDIA RTX 3090s (24 GB each), and the merged model runs inference on a single 24 GB card. The one exception is the 14B text-to-video world model in the conditioning-gradient comparison (Section 3), which we ran on an A100. A grounded, unified physical-AI model on this axis needs neither a foundation-scale pretrain nor a large compute budget; it needs the right data.

D Full tables

Per-slot and per-object tables with Wilson intervals, seed ranges, the disease-table per-model detail, the protection-capacity and coupling-dose curves, the regularizer-null sweep, and the LIBERO-Plus per-factor table are released with the code alongside the frozen per-slot result artifacts.